



KTU NOTES

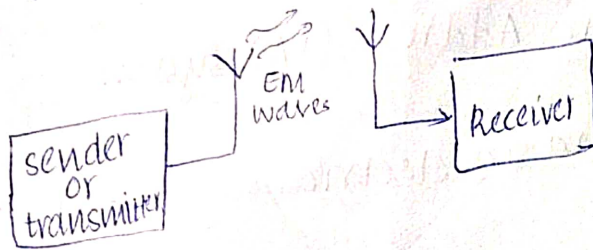
The learning companion.

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NOTIFICATIONS | SOLVED QUESTION PAPERS**

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Electronic communication



Modulation

changing amplitude, frequency, phase of carrier wave

- Base band signal will be enclosed in carrier messages ^(20 Hz - 3 kHz) are usually in low freq.

Need for modulation

- $E = h\nu$
when ν is $\downarrow$, $E \downarrow$
modulatⁿ is done to high freq.
- Radiation is efficient only at high freq.
- radiation can only be done with antennae.
if no modulation, then λ will be $\uparrow$
low freq. is modulated with high freq.
Thus freq, amplitude, phase of carrier wave is changed to ^{that of} modulating wave

freq. change $\rightarrow$ frequency modulation

amplitude change $\rightarrow$ amplitude modulation

so if $f \uparrow$, $\lambda = \frac{c}{f}$ so λ will be less

$\lambda/4$: quarter wave antennas

antennae height should be less

all freq. lie in same band. so frequency gets mixed when different freq. is used for modulation, they will never mix so lie in diff. freq. so receiver can tune in station they wish

$$v_c = V_c \sin(\omega_c t + \phi)$$

V_c : peak value (amplitude)

ϕ : phase

v_c : voltage of carrier

Analog modulation/continuous wave modulation

2 types

Amplitude modulation (AM)

UDSF GECT

Angle modulation

(a) Frequency modulation (FM)

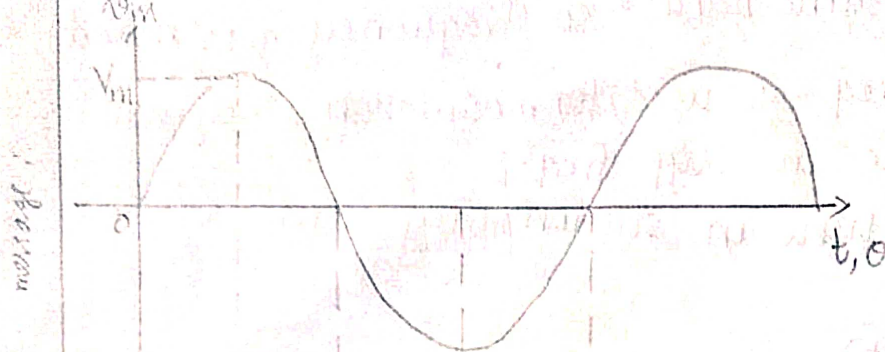
(b) Phase modulation (PM)

Digital modulation

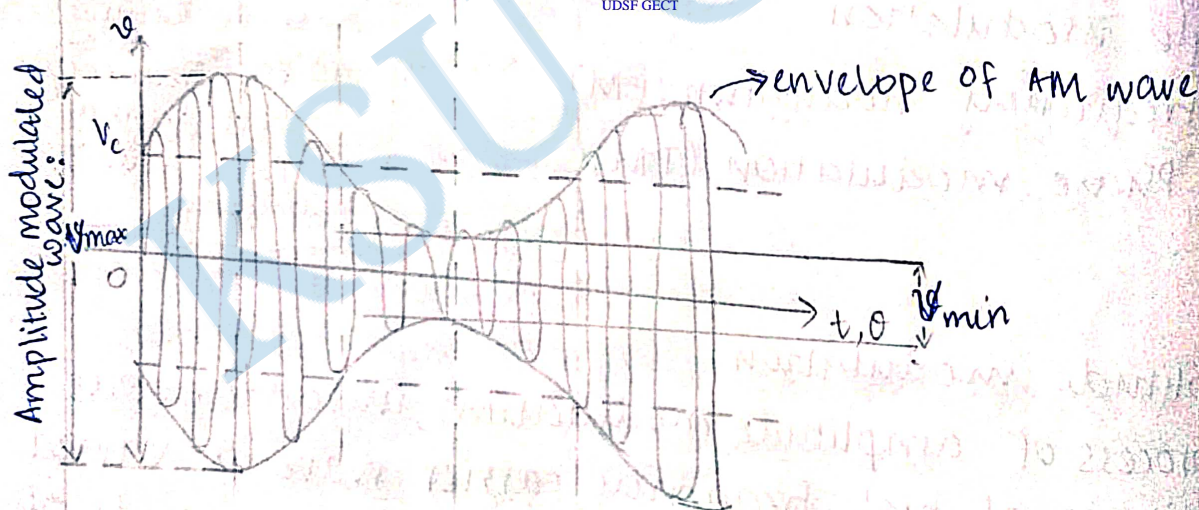
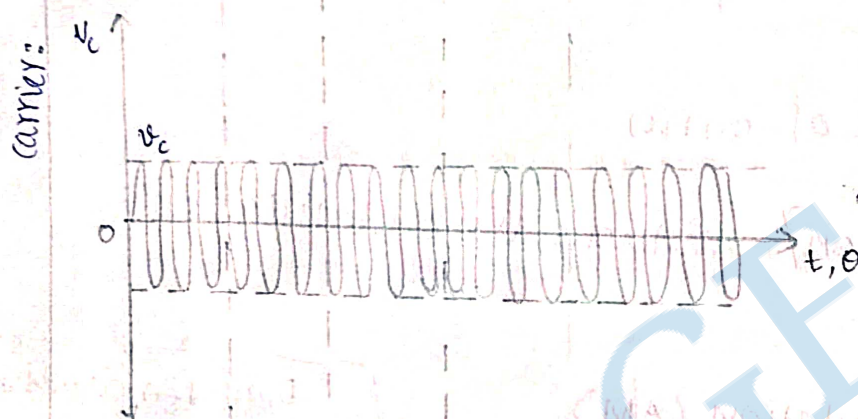
- (i) ASK: amplitude shift keying
- (ii) FSK: Frequency shift keying
- (iii) PSK: phase shift keying

Amplitude modulation

In process of amplitude modulation, instantaneous amplitude of high frequency carrier wave is varied in proportion to the instantaneous amplitude of the message voltage value, keeping frequency and phase of carrier wave a constant.



$$v_m = V_m \sin \omega_m t \quad \text{--- (1)}$$



- only instrument which display electrical waveform is CRO
- AM index : also called depth of A.M.

$$m = \frac{V_m}{V_c}$$

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

$$0 \leq m \leq 1$$

when message wave voltage is 0, $m=0$

when $m=1$, $V_m=V_c$

when $m > 1$, no signal is received by receiver

This ~~no~~ is called over modulation distortion
happens when message amplitude overlaps carrier amplitude.

$$v_m = V_m \sin \omega_m t \quad \text{---(1)}$$

$$v_c = V_c \sin \omega_c t \quad \text{---(2)}$$

AM wave is,

$$v = A \sin \omega_c t$$

where $A = V_c + V_m$

$$A = V_c + V_m \sin \omega_m t$$

$$v = (V_c + V_m \sin \omega_m t) \sin \omega_c t$$

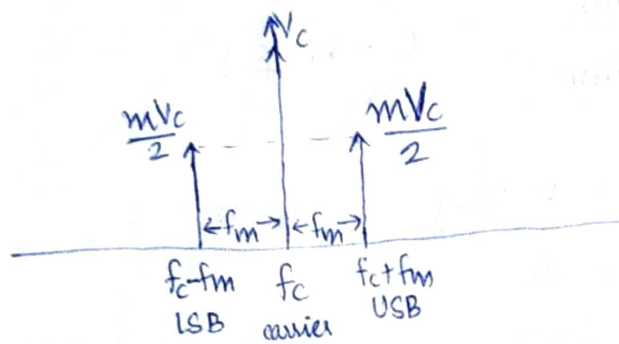
$$v = \underbrace{V_c \sin \omega_c t}_{\text{full carrier}} + V_m \sin \omega_m t \sin \omega_c t$$

$$\begin{aligned} \sin A \sin B \\ = \frac{1}{2} [\cos(A-B) - \cos(A+B)] \end{aligned}$$

$$v = V_c \sin \omega_c t + \frac{V_m}{2} \cos(\omega_c - \omega_m)t - \frac{V_m}{2} \cos(\omega_c + \omega_m)t$$

$$m = \frac{V_m}{V_c} \quad \therefore V_m = m V_c$$

$$v = \underbrace{V_c \sin \omega_c t}_{\text{Full carrier}} + \underbrace{\frac{m V_c}{2} \cos(\omega_c - \omega_m)t}_{\text{LSB}} - \underbrace{\frac{m V_c}{2} \cos(\omega_c + \omega_m)t}_{\text{USB}}$$

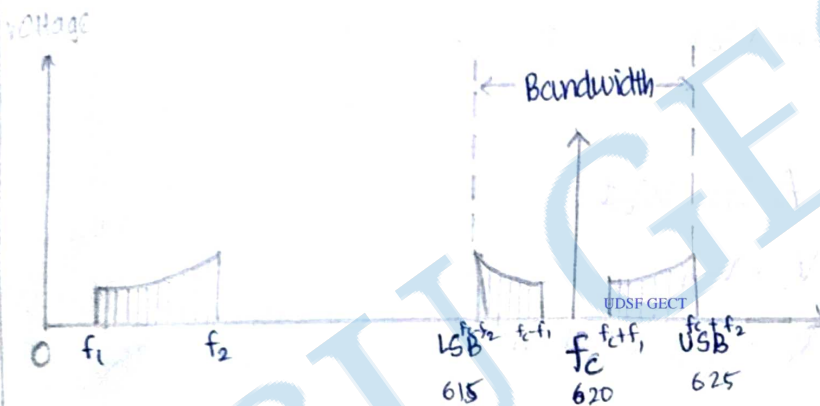


LSB: lower side band

USB: upper side band

Bandwidth of AM = $2 \times f_m$

Frequency spectrum of AM when modulated by a base band of f_1 to f_2



audio: 20 Hz - 20 kHz

Speech: 20 Hz - 3 kHz

AM radio: 20 Hz to 5 kHz
 f_1 f_2

FM radio: 20 Hz to 15 kHz

DSB-FC

Bandwidth = $2 \times f_2 = 2 \times \text{highest modulating frequency}$

- For different voltages $v_1, v_2, v_3, \dots, v_n$, modulation index will be $m_1, m_2, m_3, \dots, m_n$

Then, $m_{\text{effective}} = \sqrt{m_1^2 + m_2^2 + m_3^2 + \dots + m_n^2}$

Power in A.M.

$v = V \sin \omega t$

$$P = \frac{\left(\frac{V}{\sqrt{2}}\right)^2}{R}$$

$$P_t = P_c + P_{LSB} + P_{USB}$$

$$P_c = \frac{\left(\frac{V_c}{\sqrt{2}}\right)^2}{R} = \frac{V_c^2}{2R}$$

$$P_{LSB} = \frac{\left(\frac{mV_c}{2\sqrt{2}} \right)^2}{R} = \frac{m^2 V_c^2}{8R}$$

$$P_{USB} = \frac{\left(\frac{mV_c}{2\sqrt{2}}\right)^2}{R} = \frac{m^2 V_c^2}{8R}$$

$\therefore$ Total power,

$$P_t = \frac{V_c^2}{2R} + \frac{m^2 V_c^2}{8R} + \frac{m^2 V_c^2}{8R}$$

$$P_t = \frac{V_c^2}{2R} \left[1 + \frac{m^2}{2} \right]$$

$$P_t = P_c \left[1 + \frac{m^2}{2} \right]$$

* Calculate total power transmitted from an ordinary AM radio transmitter whose carrier frequency is 800 kHz at 100 W power and modulated to a depth of

(i) 0% (ii) 10% (iii) 50% (iv) 100%.

$$\rightarrow f_c = 800 \text{ kHz}$$

$$P_c = 100 \text{ W}$$

$$(i) P_{t1} = P_c \left(1 + \frac{m^2}{2}\right) \quad \text{given } m=0$$

$$P_{t1} = 100 (1+0)$$

$$\underline{P_{t1} = 100 \text{ W}}$$

$$(ii) P_{t2} = P_c \left(1 + \frac{m^2}{2}\right) \quad m = \frac{10}{100}$$

$$= 100 \times \left(1 + \frac{1}{200}\right)$$

$$= 100 \times \frac{201}{200} = \underline{100.5 \text{ W}}$$

$$(iii) P_{t3} = P_c \left(1 + \frac{m^2}{2}\right) \quad m = \frac{50}{100} = \frac{1}{2}$$

$$= 100 \left(1 + \frac{1}{8}\right)$$

$$= 100 \times \frac{9}{8} = \underline{112.5 \text{ W}}$$

$$(iv) P_{t4} = P_c \left(1 + \frac{m^2}{2}\right) \quad m = \frac{100}{100} = 1$$

$$= 100 \left(1 + \frac{1}{2}\right)$$

$$= 100 \times \frac{3}{2} = \underline{150 \text{ W}}$$

Balanced

carrier wave will be zero

therefore, only LSB and USB will be there

DSB - SC

Here same signal is send twice.

So a filter is used.

Filter will select either LSB or USB.

This is called Single side band (SSB).

SSB has only one band.

When bandwidth is less, we can save power and we can include more channels.

Disadvantages of SSB

- High cost
- some information may be cut

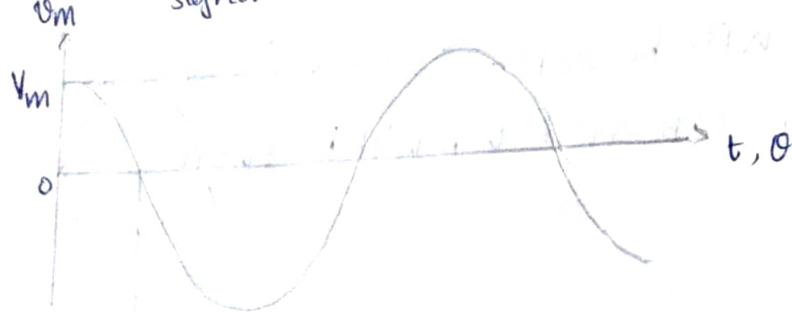
HAM: Hobby
amateurs

→ In T.V., U.S.B + a part of L.S.B is used for signal.

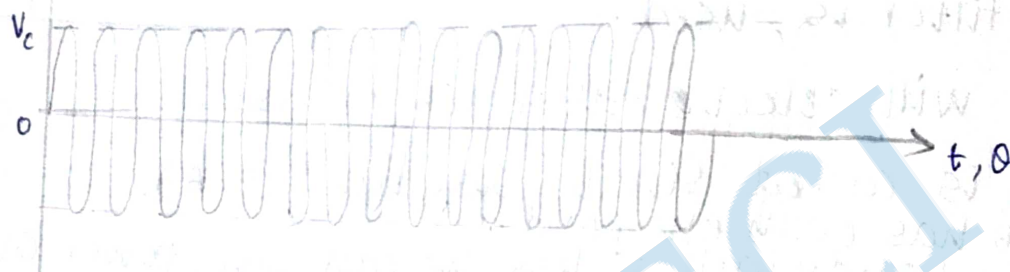
This is called vestigial side band

Frequency modulation (FM)

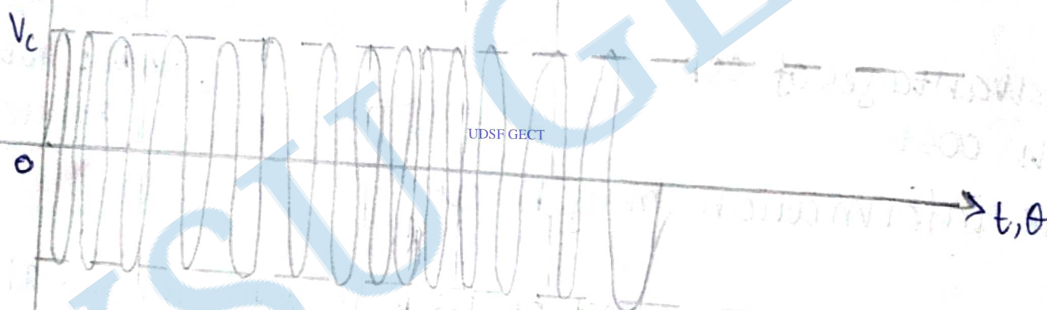
- In the process of frequency modulation, the instantaneous frequency of a high frequency carrier is varied in proportion to the instantaneous amplitude of the message/modulating voltage, keeping the amplitude and phase of the carrier a constant.



carrier



FM wave



- message signal is $v_m = V_m \cos \omega_m t$ — (i)
- carrier signal is $v_c = V_c \cos \omega_c t$ — (ii)

$f = f_c + k_f v_m f_c$; frequency deviation

- FM wave is $v = V_c \sin \theta$ — (iii)

$$\theta = \int \omega \cdot dt$$

$$\omega = \omega_c + k_f V_m \cos \omega_m t$$

$$\theta = \int [\omega_c + k\omega_c V_m \cos \omega_m t] \cdot dt$$

$$\theta = \omega_c t + \frac{k\omega_c V_m \sin \omega_m t}{\omega_m}$$

$$\theta = \omega_c t + \frac{(k f_c V_m)}{f_m} \sin \omega_m t$$

$$\theta = \omega_c t + \frac{\delta}{f_m} \sin \omega_m t \quad \text{--- (A)}$$

Maximum Frequency deviation, δ

$$\delta = k V_m f_c$$

$$m_f = \frac{\delta}{f_m}$$

$$\frac{\delta}{f_m} = \text{modulation index of FM wave}$$

$$\text{FM index} = \frac{\text{Max. freq. deviation}}{\text{modulating frequency}}$$

compare m of AM and FM

AM	FM
$m = \frac{V_m}{V_c}$	$m_f = \frac{\delta}{f_m} = \frac{k f_c V_m}{f_m}$
$0 \leq m \leq 1$	$m_f > 0$
	- If $m_f < 1$, it is called narrow band F.M. → used for point to point communication - If $m_f > 1$, wide band F.M. → used for music, radio broadcasting

$$\therefore (A) \Rightarrow$$

$$\theta = \omega_c t + m_f \sin \omega_m t$$

FM wave is,

$$v = V_c \sin(\omega_c t + m_f \sin \omega_m t)$$

Bessel function : sine of sine wave

- * A wave is defined as $v = 10 \sin(10^8 t + 8 \sin 300 t)$
 Estimate amplitude, frequency of carrier, m_f , max. freq. deviation
 and modulating frequency

→

$$v = 10 \sin(10^8 t + 8 \sin 300 t) \text{ V}$$

$$v = V_c \sin(\omega_c t + m_f \sin \omega_m t)$$

$$V_c = 10 \text{ V}$$

$$\omega_c = 10^8$$

$$\Rightarrow f_c = \frac{10^8}{2\pi} = 1.59 \times 10^7 \text{ Hz} = \underline{\underline{15.9 \text{ MHz}}}$$

$$m_f = 8$$

$$\frac{\delta}{f_m} = m_f$$

$$\omega_m = 300$$

$$\delta = m_f \times f_m$$

$$f_m = \frac{300}{2\pi} = \underline{\underline{47.7 \text{ Hz}}}$$

$$\frac{\delta}{f_m} = m_f$$

$$\delta = 47.7 \text{ Hz} \times 8 = 381.6 \text{ Hz}$$

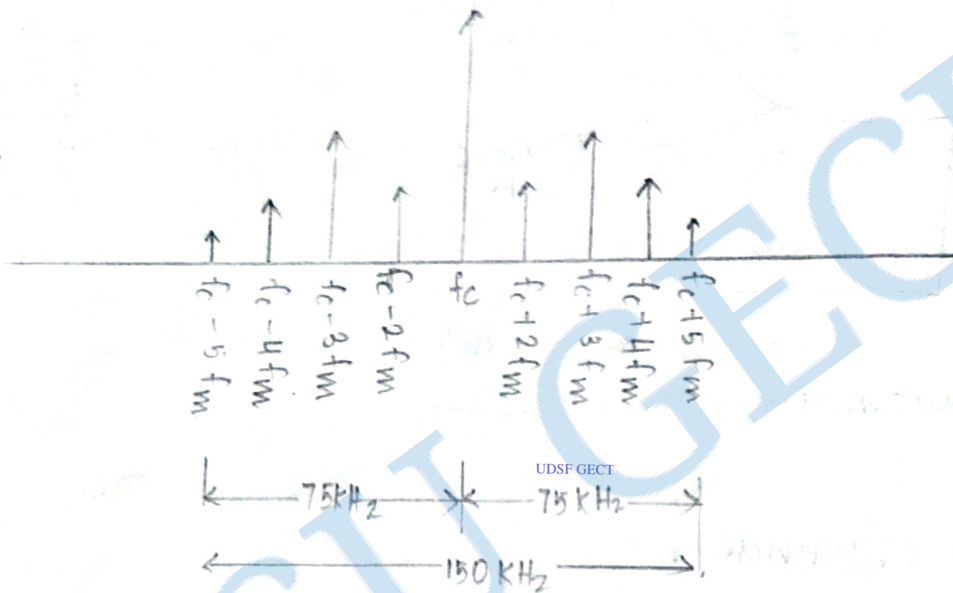
$$v = V_c \sin(\omega_c t + \dots)$$

Bessel function,

$$= I_0 \sin \omega_c t \pm I_1 \sin (\omega_c \pm \omega_m) t \\ \pm I_2 \sin (\omega_c \pm 2\omega_m) t \\ \pm I_3 \sin (\omega_c \pm 3\omega_m) t$$

$I_0, I_1 \dots \Rightarrow$ Jacobians.

varies for all m_f, f_m



Comparison of FM and AM

- FM $\Rightarrow$ highly quality because FM is highly noise immune (amplitude is constant $\Rightarrow$ noise is clipped)
 - $\Rightarrow$ carrier wave is of high frequency range, thus, noise do not get in it
 - $\Rightarrow$ all power is useful $\Rightarrow$ signals can be taken out
 - $\Rightarrow$ Frequency re-usage $\Rightarrow$ FM Freq. range 30 MHz - 300 MHz
- Frequency do not go out of LOS (line of sight), thus frequencies won't interfere

∴ same frequencies can be FM (reused) without interference
 FM: Fidelity ↑ → quality high

Limitation:

high bandwidth

∴ amplifying circuits are complicated

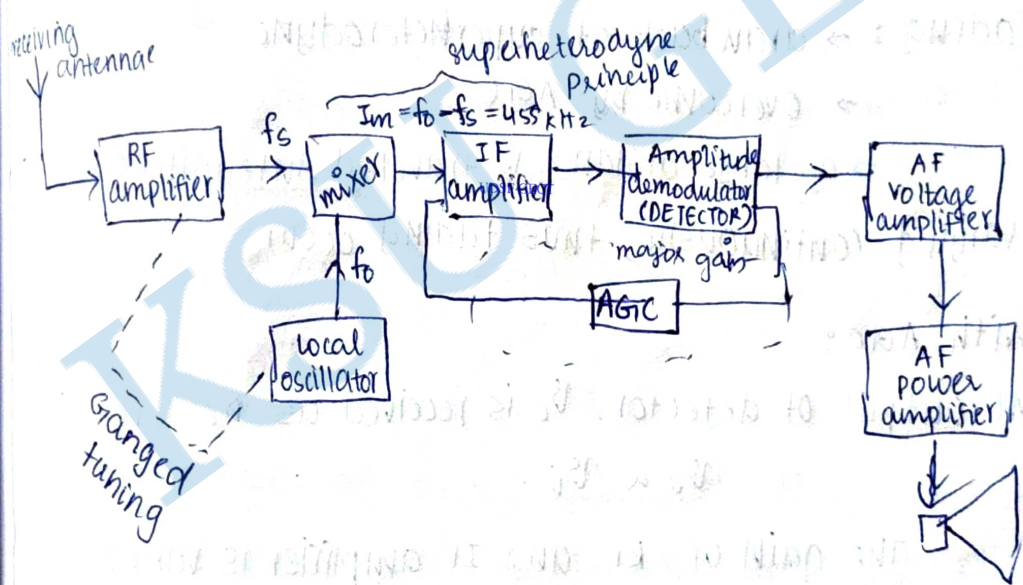
∴ stations ↓

space wave propagation: Range of distance from antenna to station



space wave propagation

Super heterodyne AM radio receiver (mix)



Receiving antennae based on principle of Faraday's law of EMI

AM → medium wave 620 - 1650 kHz
 short wave 1650 kHz - 3 MHz } Band

10 kHz : bandwidth

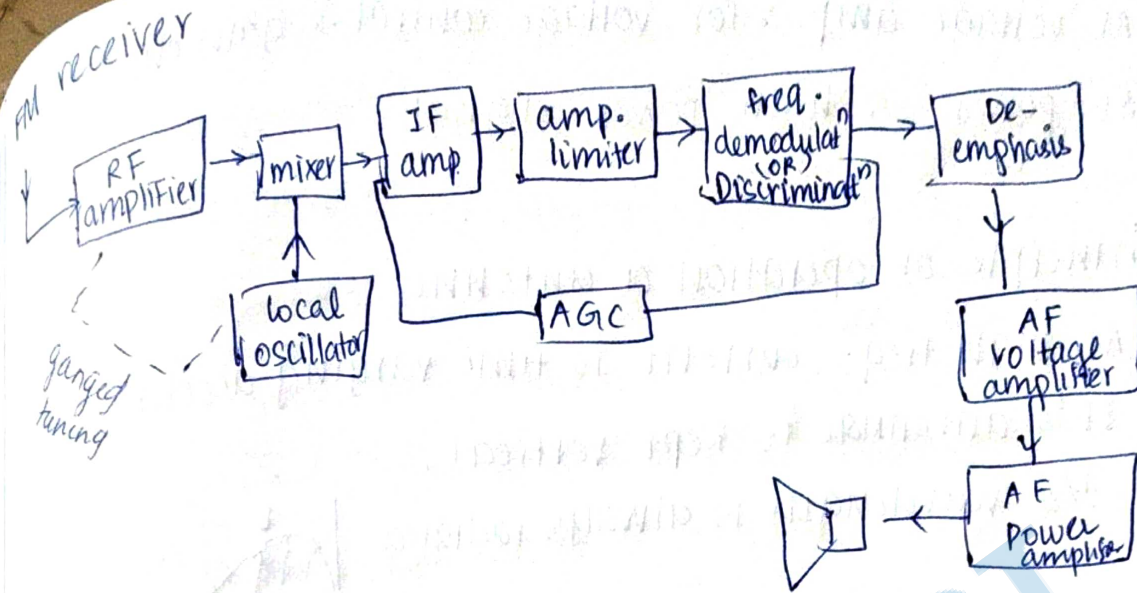
FM → 88 MHz - 108 MHz; band

150 kHz : bandwidth

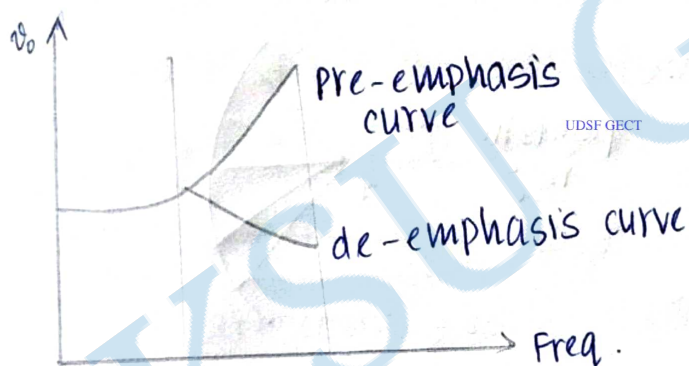
- Mixer (local oscillator): $f_m = f_o - f_s \Rightarrow$ Intermediate Frequency
 $f_o - f_s$ is constant
 AM $\rightarrow f_o - f_s = 455 \text{ KHz}$
 FM $\rightarrow f_o - f_s = 10.7 \text{ MHz}$
- Audio frequency voltage amplifier $\rightarrow$ gain in this stage changes (voltage)

$$V_o = A V_i \Rightarrow \text{if fluctuates, then output also varies}$$
 $\therefore$ as $V_i \uparrow$, A should be decreased $\Rightarrow V_o$ become steady
 This reducing A is called Automatic gain control [AGC]
- Fading: $\rightarrow$ drawback of superheterodyne
 $\rightarrow$ overcome by AGC
 $\rightarrow$ as large distance is travelled, intensities are varying continuously, thus fading occur
- With AGC:
 At output of detector, V_o is received as DC level

$$V_o \propto V_i$$
 $\therefore$ The gain of RF and IF amplifier is varied
 $\rightarrow$ as $V_i \uparrow$, A is $\downarrow$
 $\Rightarrow$ automatic gain control
- AGC produce control signal that control gain of IF amplifier (also RF amplifier)
 Thus controlling V_o of detector/discriminator



De-emphasis $\Rightarrow$ smaller gain at receiver
 $\Rightarrow$ actual signal can be taken



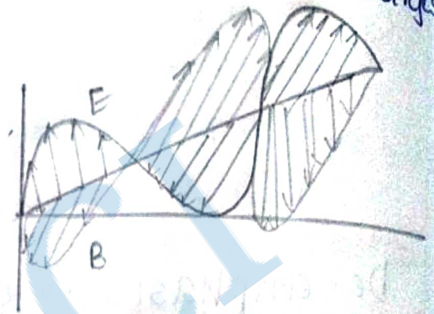
- Antennae : EMF induced $\rightarrow$ current produced
- RF amp. : selects freq. (150 kHz) and amplify if other frequency gets it, it is tuned
- Mixer : Sender frequencies are mixed with a local freq.

$$f_b - f_s = 1\text{M} = 10.7\text{MHz}$$
- Amplitude limiter : extra noises are taken out with high amplitude
- Discriminator : audio freq. are separated, gain is made strong

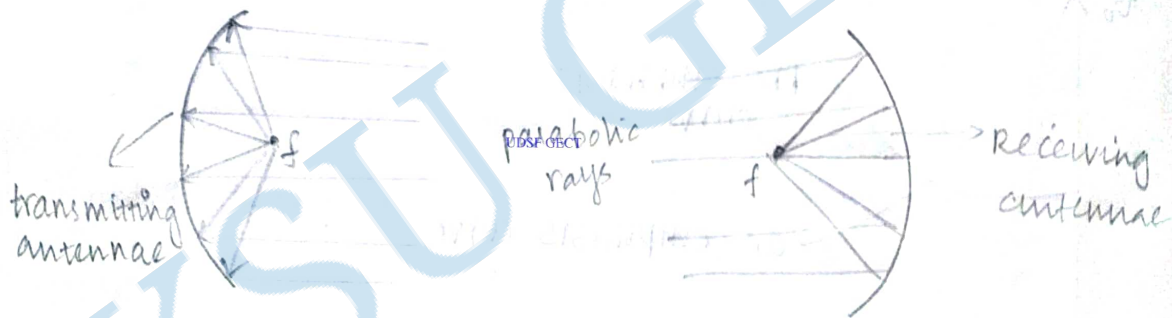
- AF voltage amp. : for voltage control $\rightarrow$ gain changes
- AF power : high power signal

Principle of operation of antenna

- At high freq., current is time varying accelerated charges
- It's antennae is kept vertical,
- $\lambda/4$ wavelength is always radiated



Parabolic reflector (Dish)



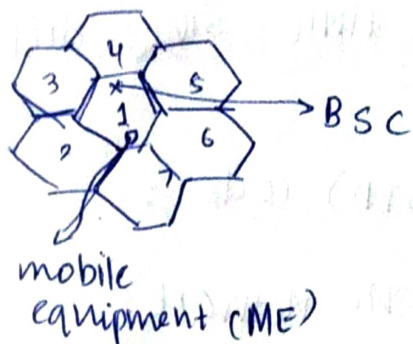
Applications

- microwave tower
- telephone
- television
- DTH (direct to home)
- Wireless LAN and WAN
- radio telescope (weather forecast)
- RADAR
- To navigate

GSM and mobile communication

GSM : global system for mobile communication

there is a base station controller (BSC) at each centre



MSC : mobile switching Centre

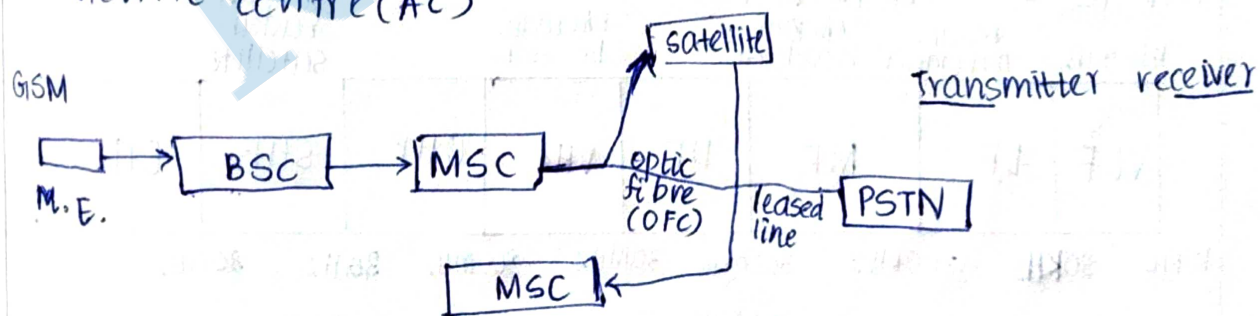
SSM : subscriber identify module : it is unique for each mobile
code from ME $\rightarrow$ BSC $\rightarrow$ threshold limit
freq. band is fixed for each service provider

VLR : visiting location register

HLR : Home location register

All BCS's are connected to an MSC

Authentic centre (AC)



carrier frequency is allotted by BSC.

PN code : Pseudo random noise

CDMA : code deficient multiple access

Here, many signals can be passed ^{in same medium} without any interference

→ ~~eg~~ Time division multiple access (TDMA)

→ Frequency " " " (FDMA) ; many freq. can be passed at same time. Since diff. freq, they do not mix

- Subscriber ~~trunk~~ dialling (STD) code :
- PSTN : Public switched telephone network

Advantages

- only communication when persons are in motion
- SMS, MMS, videos, pictures can be shared
- Teleconference, videoconference

Drawbacks

- emit radiations which is hazardous

Range of frequencies

Telegraphy	marine navigation	AM radio broadcast		FM radio broadcast		microwave radar satellite	
VLF	LF	MF	HF	VHF	UHF	SHF	EHF
10kHz	30kHz	300kHz	3000kHz (or) 3MHz	30MHz	300MHz	3GHz	30GHz

VLF : very low frequency

LF : low frequency

MF : medium frequency

HF : high frequency (for long distance communication)

• - Morse code

above 1 MHz : sky wave propagation

VHF : Very high Frequency

FM radio broadcast : 88 MHz - 108 MHz

UHF : Ultra high frequency

SHF : Super high frequency

EHF : Extra high frequency

Bands and ranges

L band $\rightarrow$ 1 GHz - 2 GHz (satellite, mobile, radar, microwave)

S band $\rightarrow$ 2 to 4 GHz

C band $\rightarrow$ 4 to 8 GHz

X band $\rightarrow$ 8 to 12 GHz

Ku band $\rightarrow$ 12 to 18 GHz

K band $\rightarrow$ 18 to 27 GHz

Ka band $\rightarrow$ 27 to 40 GHz

Ku : under K

Ka : above K

Block diagram of an electronic instrumentation system

Transducer : convert energy of one form to electrical energy

